Co-located Multi-user VR System
We Are Headset

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***Abstract*—¡¡¡¡¡¡¡ HEAD Co-located multi-user virtual reality (VR) training enables participants to share physical space while collaborating in virtual environments, yet its technical feasibility using consumer hardware remains under-validated. This study evaluates a 3-user co-located system using Meta Quest 3 wireless headsets against evidence-based technical benchmarks relevant to professional training. We measured network latency, spatial calibration precision and stationary drift relative to a shared anchor, frame rate stability against the 72Hz target, and device thermals via battery temperature during extended runs. The system met the selected benchmarks in these proxy measures throughout the test sessions, supporting the feasibility of consumer wireless VR for co-located training scenarios at substantially lower cost than enterprise solutions. ======= Co-located multi-user virtual reality (VR) training enables participants to share physical space while collaborating in virtual environments, yet its technical feasibility using consumer hardware remains under-validated. This study evaluates a 3-user co-located system using Meta Quest 3 wireless headsets against evidence-based technical benchmarks relevant to professional training. We measured network latency, spatial calibration precision and stationary drift relative to a shared anchor, frame rate stability against the 72Hz target, and device thermals via battery temperature during extended runs. The system met the selected benchmarks in these proxy measures throughout the test sessions, supporting the feasibility of consumer wireless VR for co-located training scenarios at substantially lower cost than enterprise solutions. ¿¿¿¿¿¿¿ 32df077fa8bbb82b5374f4992fe0114abd8167f7**

***Index Terms*—Virtual Reality, Co-located VR, Multi-user Systems, Collaborative Training, Meta Quest 3, Wireless VR, Spatial Calibration, Professional Training, Consumer VR Hardware**

I. Introduction

Virtual Reality (VR) training systems have demonstrated significant potential for professional education, particularly in safety-critical domains such as emergency medical services [1], maintenance operations [2], and crisis response [3]. ¡¡¡¡¡¡¡ HEAD However, most existing multi-user VR solutions address scenarios where users connect from geographically distributed locations [4], leaving a critical gap in systems designed for co-located training where multiple users share the same physical space. This distinction is not merely technical since co-located training enables real-world teamwork dynamics, immediate physical assistance between trainees, and authentic spatial coordination that remote systems cannot replicate. Standalone wireless VR headsets, particularly the Meta Quest 3, present opportunities for co-located multi-user training, eliminating cable-related usability issues that caused 66.7% of users to rate tethered systems poorly [1]. However, co-located systems introduce unique challenges achieving calibration accuracy to prevent user collisions [5], maintaining low network latency [4], providing spatial awareness interfaces [6], and ensuring visual consistency across headsets [7]. While enterprise VR rooms, such as Virtualware's VIROO demonstrate commercial viability, their high costs ($50,000+) limit accessibility. This work addresses the need for an affordable, portable co-located VR system using Meta Quest 3 headsets (≈$500 each), targeting co-location requirements including collision prevention [5], spatial awareness [6], and high-precision avatar alignment.

A. Motivation and Significance

Professional training in safety-critical domains demands realistic, repeatable, and scalable practice. Traditional high-fidelity training rigs are costly and logistically heavy, limiting frequent practice and team-based drills [3]. Consumer wireless headsets, such as Meta Quest 3 promise large cost reduction, but require validation against evidence based technical benchmarks: precise calibration (<10mm) for collision avoidance [5], low end-to-end latency to preserve shared action timing [4], and spatial awareness interfaces to support multi-user coordination [6]. Visual consistency across participants is also critical as mismatches degrade collaborative performance [7]. This work addresses the technical foundation for such validation by implementing a 3-user co-located prototype that demonstrates automated spatial anchor colocation, performance monitoring infrastructure, and repeatable multi-device coordination over local WiFi. The system provides a reference implementation and establishes a metrics collection framework for future empirical studies to measure learning, safety, and usability outcomes with actual participants. ======= However, most existing multi-user VR solutions address scenarios where users connect from geographically distributed locations [4], leaving a critical gap in systems designed for colocated training where multiple users share the same physical space. This distinction is not merely technical—co-located training enables real-world teamwork dynamics, immediate physical assistance between trainees, and authentic spatial coordination that remote systems cannot replicate. Standalone wireless VR headsets, particularly the Meta Quest 3, present opportunities for co-located multi-user training, eliminating cable-related usability issues that caused 66.7% of users to rate tethered systems poorly [1]. However, co-located systems introduce unique challenges achieving calibration accuracy to prevent user collisions [5], maintaining low network latency [4], providing spatial awareness interfaces [6], and ensuring visual consistency across headsets [7]. While enterprise VR rooms (e.g., Virtualware's VIROO) demonstrate commercial viability, their high costs ($50,000+) limit accessibility. This work addresses the need for an affordable, portable co-located VR system using Meta Quest 3 headsets (≈$300 each), targeting co-location requirements including collision prevention [5], spatial awareness [6], and high-precision avatar alignment.

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C. Research Questions

This prototype implementation addresses key technical challenges identified in co-located multi-user VR literature. The system design targets integration of validated requirements from systematic review to create a foundation for future empirical research.

¡¡¡¡¡¡¡ HEAD ======= Research Questions: ¿¿¿¿¿¿¿ 32df077fa8bbb82b5374f4992fe0114abd8167f7

- **RQ1**: How can automated spatial colocation be implemented for a 3-user configuration using Meta's OVR Colocation Discovery API with shared spatial anchor-based alignment? [**?**], [**?**]
- **RQ2**: How accurately and stably can the shared coordinate system be maintained during extended sessions, as measured by calibration error and drift relative to the shared anchor?
- **RQ3**: What on-device monitoring infrastructure is required to continuously log key technical metrics (network latency, frame rate, calibration error, and thermals) during long-duration tests?
- **RQ4**: What device setup and deployment workflow enables reproducible multi-headset development and experimentation?
- **RQ5**: What are the practical feasibility boundaries of a consumer, wireless 3-user co-located Quest 3 system under a dynamic training proxy, in terms of performance, stability, and operational limitations? ¡¡¡¡¡¡¡ HEAD

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II. State of the Art

A. Collaborative VR for training and industrial work

¡¡¡¡¡¡¡ HEAD Multi-user VR has been studied as a training and collaboration medium in safety-critical contexts. In paramedic training, Schild et al. [1] found usability issues to be a limiting factor, with 66.7% of participants rating tethered systems below the acceptability threshold on the System Usability Scale (SUS < 70), and reported strong relationships between usability and presence, such as experienced realism $r = 0.73$, $p < 0.001$. In industrial maintenance, Heinonen et al. [2] investigated collaborative VR for documentation review and risk assessment with domain experts (n=9) and reported positive ratings and improved spatial understanding, highlighting that practical value depends on task fidelity and usable review tooling.

B. Network latency and multi-user QoE

Network responsiveness is a key determinant of collaboration quality in interactive multi-user VR. Van Damme et al. [4] showed that latency strongly influences collaborative quality of experience (QoE): $\leq$ 75ms round-trip time (RTT) supports good collaboration, while $>$ 300ms causes severe degradation. Importantly, they note that multi-user experiences are not only affected by uniform delay but also by bursty network behavior, and that subjective perception is shaped by user role, task type, and movement rather than objective performance alone.

C. Co-location calibration for SLAM-tracked headsets

For co-located VR, correct alignment between physical and virtual relative user positions is safety-critical. Reimer et al. [5] evaluated calibration methods for headsets tracked using Simultaneous Localization and Mapping (SLAM) without external tracking, including fixed-point, marker-based, and hand-tracking-based methods. They reported that hand tracking-based calibration provided the best accuracy and consistency (on the order of $\sim$10mm) and reduced infrastructure requirements. However, their evaluation was limited to two Oculus Quest devices and emphasized one-time calibration, leaving multi-device validation on newer headsets and long-duration stability under realistic motion as open questions.

D. Spatial awareness and cross-user consistency

Beyond geometric alignment and latency, collaborative effectiveness depends on users maintaining awareness of teammates and sharing consistent representations. Chen et al. [6] studied overview-based interfaces for collaborative VR (n=36), showing that interactive World-in-Miniature (WIM) and 2D map interfaces can improve task performance, usability, and social presence while reducing VR sickness; WIM was particularly beneficial in higher task complexity. Separately, Weiss et al. [7] demonstrated that visual-haptic mismatches and, crucially, mismatched visualizations between collaborators can degrade both performance and experience in co-located collaborative tasks, implying that symmetric, consistent representations across headsets matter for robust collaboration.

E. Summary of Research Gap

While existing literature establishes key baselines for collaborative feasibility, such as RTT limits for collaboration [4] and $\sim$10mm calibration targets for collision risk reduction [5], evidence is fragmented across different hardware generations, user counts, and experimental contexts. Prior work often studies networking (typically two-user collaboration with controlled latency) separately from co-location calibration (often two devices and one-time procedures) and from interface support for awareness (often task-focused lab studies). Consequently, there remains limited validation of whether a consumer, standalone 3-user system can simultaneously (i) establish shared spatial alignment, (ii) maintain it over extended durations, and (iii) meet collaboration-relevant performance targets while using consistent representations across devices. This work addresses that intersection by integrating co-location, networking, and continuous on-device measurement into a unified 3-user Quest 3 testbed. ======= Multi-user VR has been studied as a training and collaboration medium in safety-critical contexts. In paramedic training, Schild et al. [1] found usability issues to be a limiting factor (66.7% rated tethered systems poorly; SUS < 70) and reported strong relationships between usability and presence (e.g., experienced realism $r = 0.73$, $p < 0.001$). In industrial maintenance, Heinonen et al. [2] investigated collaborative VR for documentation review and risk assessment with domain experts (n=9) and reported positive ratings and improved spatial understanding, highlighting that practical value depends on task fidelity and usable review tooling.

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Beyond geometric alignment and latency, collaborative effectiveness depends on users maintaining awareness of teammates and sharing consistent representations. Chen et al. [6] studied exocentric perspective interfaces for collaborative VR (n=36), showing that interactive World-in-Miniature (WIM) and 2D map interfaces can improve task performance, usability, and social presence while reducing VR sickness; WIM was particularly beneficial in higher task complexity. Separately, Weiss et al. [7] demonstrated that visuo-haptic mismatches and, crucially, mismatched visualizations between collaborators can degrade both performance and experience in co-located collaborative tasks, implying that symmetric, consistent representations across headsets matter for robust collaboration.

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III. Approach

This study investigates a co-located multi-user VR configuration using wireless headsets to address the gap between distributed VR collaboration systems and the requirements of shared physical training environments. The approach is intentionally scoped as a *technical feasibility validation* of a consumer 3-user setup, rather than a learning-outcome study.

A. How the Approach Differs

¡¡¡¡¡¡¡ HEAD Prior co-located systems commonly rely on external tracking infrastructure, such as multi-camera setups or manual calibration workflows, while multi-user networking studies frequently focus on two-user collaboration with controlled, simulated network conditions. Relative to these solution families, our approach integrates (i) *anchor-based automated co-location* using inside-out tracked consumer headsets, (ii) *local rendering with lightweight synchronization* over WiFi rather than server-rendered streaming architectures with throughput/latency requirements, and (iii) *continuous on-device instrumentation* so performance and stability can be evaluated during long duration sessions.

The approach is designed to be *better for validating feasibility in practice* because it uses the same constraints a training deployment would face (standalone headsets, consumer WiFi, minimal setup time), while tying acceptance criteria to evidence-based thresholds from the literature: safe alignment accuracy for collision risk reduction [5] and collaboration-relevant RTT targets [4]. Finally, by standardizing multi-headset setup and APK deployment through Meta Quest Developer Hub (MQDH) [**?**], the workflow reduces per-device variance and supports repeatable iteration across all three headsets.

While our prototype does not implement a dedicated spatial awareness UI like WIM, it emphasizes representation consistency: the same anchored scene content is rendered on each headset. Prior work has shown that mismatched spatial representations across users can reduce collaborative performance. [7]. ======= Prior co-located systems commonly rely on external tracking infrastructure (e.g., multi-camera setups) or manual calibration workflows, while multi-user networking studies frequently focus on two-user collaboration with controlled, artificial network impairments. Relative to these solution families, our approach integrates (i) *anchor-based automated co-location* using inside-out tracked consumer headsets, (ii) *local rendering with lightweight synchronization* over WiFi rather than server-rendered streaming architectures with stringent throughput/latency requirements, and (iii) *continuous on-device instrumentation* so performance and stability can be evaluated during long-duration sessions.

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While our prototype does not implement a dedicated spatial awareness UI (e.g., WIM), it emphasizes representation consistency: the same anchored scene content is rendered on each headset, motivated by findings that mismatched representations can degrade collaborative performance [7]. ¿¿¿¿¿¿¿ 32df077fa8bbb82b5374f4992fe0114abd8167f7

B. Design Targets (Benchmarks)

The feasibility validation uses literature-derived targets as benchmark criteria:

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- **Collaboration latency**: Maintain $\leq$ 75ms RTT as a target threshold for good collaborative QoE [4].
- **Co-location accuracy**: Keep calibration error below 10mm as a conservative, safety-relevant reference for collision prevention [5].
- **Visual performance**: Maintain stable rendering at the device target refresh rate (72Hz) to support comfort and consistent interaction.
- **Operational stability**: Meet the above targets over extended continuous sessions (tens of minutes to $\sim$1.5 hours) without external instrumentation.

C. System Overview

The experimental system consists of three Meta Quest 3 standalone headsets operating within a shared physical space and networked via WiFi. Co-location is established using Meta's OVR Colocation Discovery API [**?**] with a shared spatial anchor [**?**], while multi-user state synchronization is implemented using Photon Fusion 2 [**?**], [**?**]. Implementation details and the concrete Unity components are described in Section IV-D and in the *Experimental Prototype* section.

D. Research Methodology

The prototype follows a technical demonstration approach to validate whether the integrated system meets the benchmarks in a realistic runtime proxy. Each headset logs key metrics at 1Hz (Section IV-F), enabling longitudinal analysis of latency, frame rate, calibration error, and thermals without external measurement tools. Long-duration sessions (60+ minutes) are used to surface stability issues that short lab trials may miss, such as drift accumulation, network spikes, or thermal throttling. =======

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E. System Design

The experimental system consists of three Meta Quest 3 standalone headsets operating within a shared physical space, networked via WiFi infrastructure.

1) Hardware Configuration: The hardware setup comprises three Meta Quest 3 standalone headsets, which utilize inside-out optical tracking with hand tracking capabilities. Networking is handled by a WiFi router configured with a dedicated 5GHz channel to support low-latency local connectivity.

2) Software Architecture: The system is built on Unity 6000.0.62f1 and integrates the Meta XR SDK v81.0.0 for core VR and passthrough functionality. The Universal Render Pipeline (URP) is employed for performance-optimized rendering. Additionally, a custom `MetricsLogger` component is included for 1Hz performance monitoring, tracking FPS, RTT, calibration error, and thermals.

3) Calibration Protocol: To achieve precise co-location, the system employs an automated spatial anchor alignment procedure. The host device creates and shares a *Spatial Anchor*, a persistent, world-locked coordinate reference point generated by the headset's simultaneous localization and mapping (SLAM) system via the OVR Colocation Discovery API.

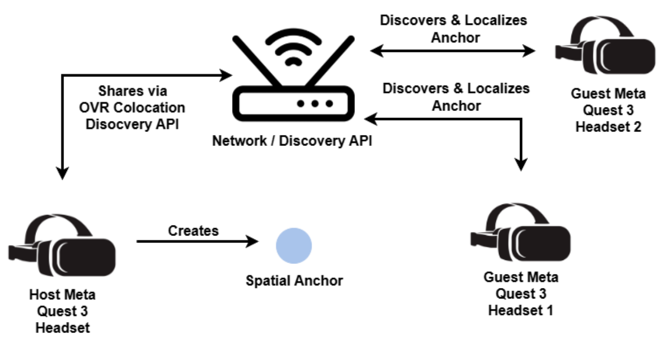

Fig. 1. Illustration of the shared physical space and spatial anchor synchronization process. The host creates an anchor which is shared via the OVR Colocation Discovery API, allowing guest headsets to align their coordinate systems.

Guest devices then discover and localize this shared anchor. Subsequently, the `ColocationManager` aligns each user's camera rig to the anchor's transform. Throughout the session, alignment error is continuously calculated as the Euclidean distance from the anchor position.

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G. Assumptions and Limitations

Several assumptions and limitations guide this study. Hardware constraints inherent to mobile processing necessitate trade-offs in visual fidelity to maintain target frame rates. Reliable WiFi connectivity is assumed, with measured RTT reflecting the quality of the local network and its contention. Finally, while co-location is established via shared anchors, the strongest safety claims require motion-robust evaluation; therefore, drift measurements under stationary conditions should be interpreted as a best-case baseline (with motion-robustness treated as future work).

With the methodological framework and design constraints established, the following section details the specific implementation of the experimental testbed used to validate these parameters.

IV. Experimental Prototype

¡¡¡¡¡¡¡ HEAD The implementation of the 3-user co-located VR training system, specifically the "Shooting Game" scenario developed for this study. Designed as a representative proxy for professional collaborative training tasks, the application places three co-located users in a shared virtual arena where they must coordinate to neutralize waves of aerial drones. This scenario was chosen to place significant demands on the system's networking and spatial alignment capabilities. ======= The implementation of the 3-user co-located VR training system, specifically the "Shooting Game" scenario developed for this study. Designed as a representative proxy for professional collaborative training tasks, the application places three co-located users in a shared virtual arena where they must coordinate to neutralize waves of aerial drones. This scenario was chosen to impose significant demands on the system's networking and spatial alignment capabilities. ¿¿¿¿¿¿¿ 32df077fa8bbb82b5374f4992fe0114abd8167f7

A. Hardware Configuration

The experimental testbed consists of three Meta Quest 3 standalone headsets, utilizing inside-out optical tracking and Meta Quest controllers. The network infrastructure is supported by a WiFi router with a dedicated 5GHz channel to ensure a low-latency local session over WiFi.

B. Software Architecture

The system is built on Unity 6000.0.62f1 using the Universal Render Pipeline (URP). The architecture leverages Meta's "Building Blocks" for rapid development of complex MR features.

1) Key Components: The *ShootingGame* scene integrates several core components (see Fig. 2). The **Colocation** system handles the spatial alignment of multiple users by utilizing Meta's Shared Spatial Anchors [**?**] to establish a common coordinate system. The **Network Manager** oversees the local session, while the **Shooting Game Manager** orchestrates the game logic and a **Spawn Manager** handles the dynamic spawning of targets (drones).

¡¡¡¡¡¡¡ HEAD Networking and shared object synchronization are implemented using Photon Fusion 2 [**?**], [**?**], enabling replicated avatars and gameplay objects through `NetworkBehaviour` components and RPC messaging.

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C. Scene Implementation: Shooting Game

The primary experimental scenario for the experiment is a Shooting Game, designed to test co-location accuracy and network latency in a dynamic, interactive environment.

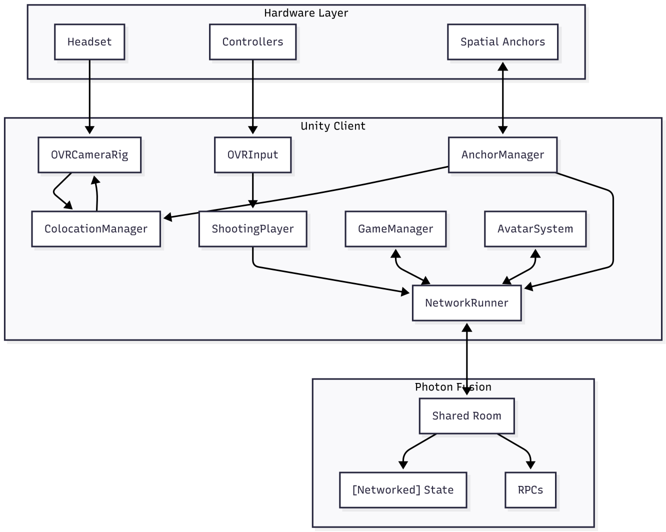


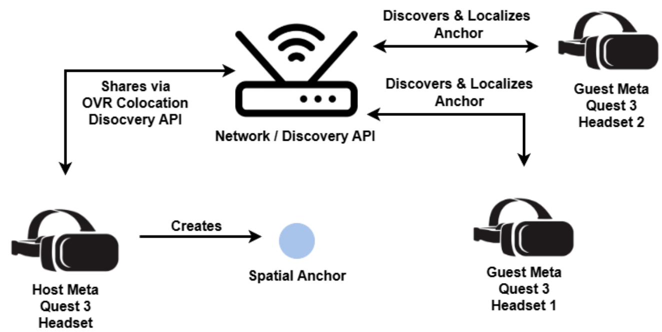
Fig. 2. System Architecture Diagram illustrating the data flow between Unity components, the networking layer, and the physical hardware layer.

1) Arena Setup: The virtual environment consists of a central `Arena` structure that serves as the shared physical reference point. Users are physically co-located around this virtual arena, allowing them to see and interact with the same virtual targets from different physical perspectives.

2) Gameplay Mechanics: ¡¡¡¡¡¡¡ HEAD Players collaborate to shoot drone targets spawned by the *Spawn Manager*. The *Shooting Game Manager* coordinates the round loop for spawning waves, tracking progression, and restart conditions. This creates real-time rendering and interaction load while preserving shared spatial alignment across headsets. ======= Players collaborate to shoot drone targets spawned by the *Spawn Manager*. The *Shooting Game Manager* coordinates the round loop for spawning waves, tracking progression, and restart conditions. This imposes real-time rendering and interaction load while preserving shared spatial alignment across headsets. ¿¿¿¿¿¿¿ 32df077fa8bbb82b5374f4992fe0114abd8167f7

D. Spatial Colocation

¡¡¡¡¡¡¡ HEAD The system established a shared coordinate system across three Meta Quest 3 headsets by creating a host-defined spatial anchor and distributing it via the OVR Colocation Discovery API [?]. In our implementation, the host advertises a local session (producing a group UUID) and places the anchor at the host headset's current position projected to the ground plane. Guest headsets perform nearby session discovery, load the shared anchor group, and localize the received unbound anchor before binding it to a scene object. Once localized, each guest aligns its tracking space/camera rig to the anchor transform, resulting in a consistent placement of the arena and interactive content across devices. For operational robustness, the implementation includes explicit discovery timing/timeout handling and logs the discovery and localization durations for debugging and repeatability.



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1) Game Loop: The game follows a simple, repeatable loop designed for continuous data collection. In the **Lobby Phase**, users join the shared session and align their coordinate systems. During the **Round Start**, the host initiates a round, spawning a wave of drone targets. In the **Active Phase**, players use their controllers to aim and shoot lasers at the drones. Finally, in the **Round End**, once all targets are cleared, scores are tallied. This cycle repeats, allowing for long-duration testing sessions.

E. Calibration Protocol

¡¡¡¡¡¡¡ HEAD The system employs an automated spatial anchor alignment procedure. First, the host device creates a spatial anchor at the center of the physical play area. This anchor is then shared via the OVR Colocation Discovery API [?] and loaded as a Shared Spatial Anchor [?]. Finally, guest devices discover the shared anchor and align their Unity coordinate system to match the host's, ensuring the arena and all game objects appear in the exact same physical location for everyone.

F. Data Collection

A custom `MetricsLogger` component is included in the scene to capture real-time performance data, including network latency (RTT) as reported by Photon Fusion [?], instantaneous and average frame rate (FPS), and calibration error measured as the Euclidean distance between the local anchor position and the shared anchor position. This logging pipeline allows sampling at 1Hz to support long-duration evaluation without external instrumentation. Each headset writes a per-session CSV file containing: session identifier, headset identifier, participant count, timestamp, frame rate, network latency (RTT), calibration error, battery temperature, battery level, and runtime state (host/client/not connected). In addition, a JSON metadata file is generated per session to capture device and runtime context (device model/name, Unity/app version, start/end time, duration, and total samples). To reduce data loss during long runs, logs are periodically flushed to disk and also persisted on application pause/quit; an optional on-headset debug overlay provides live visibility of the same key metrics during testing.

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A custom `MetricsLogger` component is included in the scene to capture real-time performance data, including network latency (RTT) to the local session host, instantaneous and average frame rate (FPS), and calibration error measured as the Euclidean distance between the local anchor position and the shared anchor position. This logging pipeline allows sampling at 1Hz to support long-duration evaluation without external instrumentation. Each headset writes a per-session CSV file containing: session identifier, headset identifier, participant count, timestamp, frame rate, network latency (RTT), calibration error, battery temperature, battery level, and runtime state (e.g., host/client/not connected). In addition, a JSON metadata file is generated per session to capture device and runtime context (device model/name, Unity/app version, start/end time, duration, and total samples). To reduce data loss during long runs, logs are periodically flushed to disk and also persisted on application pause/quit; an optional on-headset debug overlay provides live visibility of the same key metrics during testing. ¿¿¿¿¿¿¿ 32df077fa8bbb82b5374f4992fe0114abd8167f7

H. Meta Quest Device Setup and APK Deployment

To enable reproducible development and on-device testing, each Meta Quest 3 headset was configured for Android developer deployment. The setup requires enabling Developer Mode on the headset and installing the application APK through Meta Quest Developer Hub [?]. All headsets must be enrolled under a Meta developer account/organization, paired through the Meta Horizon mobile app, and configured in Developer Mode to enable USB deployment. During setup, each headset prompted for USB debugging authorization, which was accepted to allow repeat installations from the development PC. APK deployment was performed through Meta Quest Developer Hub (MQDH) by verifying the headset appeared as an authorized device, installing the build from MQDH's device manager, and launching the application from the headset library. This standardized workflow reduced per-device configuration variance and enabled reproducible build deploy cycles across the three-headset testbed.

This workflow supports repeated build deploy cycles for performance measurement sessions, while keeping the runtime environment consistent across the three headsets. ¡¡¡¡¡¡¡ HEAD

V. Results

This section presents the experimental validation of the 3-user co-located VR system during an extended co-located run. The longest continuous logging window with all three headsets active lasted 85.1 minutes (5073 samples per headset at 1Hz; 15,234 total samples across devices). The host headset additionally logged 97.0 minutes including pre-session setup and post-session. Results are organized by research question.

A. RQ1, RQ3 & RQ4: Co-location Implementation, Monitoring, and Deployment

Implementation details for colocation, logging, and deployment are described in Sections IV-D, IV-F, and IV-G. In summary, the prototype repeatedly established a shared coordinate system for three headsets using a host-advertised local session (group UUID) and a distributed shared spatial anchor, enabling consistent placement of the `Arena` and shared content. Continuous monitoring was achieved through on-device 1Hz logging (CSV + JSON metadata) across the full session duration. For reproducible multi-headset iteration, APK installation and launch were standardized using Meta Quest Developer Hub (MQDH) with Developer Mode enabled on each headset.

B. RQ2: Calibration Accuracy and Drift

Automated spatial anchor calibration achieved high accuracy under stationary baseline conditions. During the 85.1-minute 3-headset run, the client headset that logged non-zero calibration error reported a mean error of 1.21 ± 0.90mm with a maximum of 5.78mm. This remains well within Reimer et al.'s [5] $<$ 10mm safety-critical reference threshold for collision avoidance, supporting the feasibility of anchor-based co-location.

Here, *non-zero calibration error* means that the headset recorded a non-zero *residual alignment error* in our logs (i.e., a non-zero Euclidean distance between the headset's local anchor position and the shared anchor position after alignment). In this run, only one headset recorded calibration error values above zero in the CSV logs.

It should be noted that this drift evaluation was conducted with headsets placed stationary on a table. Consequently, these results represent a best-case baseline and may underestimate drift and transient misalignment that can occur under active user motion and re-localization events.

TABLE I
CALIBRATION ACCURACY (HEADSET WITH NON-ZERO LOGGING)

Metric	**Mean (mm)**	**SD**	**Max**
Error	1.21	0.90	5.78

Under these stationary conditions, calibration drift remained low over the full 85.1-minute 3-headset run, with maximum error staying well below the 10mm reference threshold.

C. RQ5: Feasibility Benchmarks and Long-Term Stability

1) Network Latency Performance: Network latency measurements show that the system's mean RTT remained below Van Damme et al.'s [4] target threshold for good collaborative quality of experience (QoE $\leq$ 75ms RTT). Over the 85.1-minute 3-headset run, mean RTT was 58.2 ± 21.3ms with a 95th percentile of 86.0ms. While the mean remains within the 75ms target, the tail latency exceeds it at P95 and includes occasional spikes (max 791.4ms) that occurred during initialization and synchronization events. =======

VI. RESULTS

This section presents the experimental validation of the 3-user co-located VR system during an extended co-located run. The longest continuous logging window with all three headsets active lasted 85.1 minutes (5073 samples per headset at 1Hz; 15,234 total samples across devices). The host headset additionally logged 97.0 minutes including pre-session setup and post-session tail time. Results are organized by research question.

A. RQ1: Technical Benchmark Achievement

1) Network Latency Performance: Network latency measurements show that the system's *mean* RTT remained below Van Damme et al.'s [4] target threshold for good collaborative quality of experience (QoE $\leq$ 75ms RTT). Over the 85.1-minute 3-headset run, mean RTT was 58.2 ± 21.3ms with a 95th percentile of 86.0ms. While the mean remains within the 75ms target, the tail latency exceeds it at P95 and includes occasional spikes (max 791.4ms) during initialization and synchronization events. ¿¿¿¿¿¿¿ 32df077fa8bbb82b5374f4992fe0114abd8167f7

¡¡¡¡¡¡¡ HEAD Despite occasional outliers, the system consistently maintained performance below the 75ms threshold, supporting the feasibility of consumer wireless networking for

TABLE II
NETWORK LATENCY PERFORMANCE

¡¡¡¡¡¡¡ HEAD **Metric** ======= **Metric**	**Mean (ms)** **Mean (ms)**	**S** **S**
¿¿¿¿¿¿¿ 32df077fa8bbb82b5374f4992fe0114abd8167f7 heightOverall	58.2	21

co-located collaborative training scenarios when supported by robust error handling.

2) Frame Rate Stability: Frame rate performance was evaluated against the Meta Quest 3's 72Hz refresh rate target set for this study. The system achieved a mean frame rate of 72.2 ± 0.7fps. ======= Despite occasional outliers, the system consistently maintained performance below the 75ms threshold, validating consumer wireless networking for safety-critical training applications when supported by robust error handling.

3) Frame Rate Stability: Frame rate performance was evaluated against the Meta Quest 3's native 72Hz refresh rate target. The system achieved a mean frame rate of 72.2 ± 0.7fps, remaining tightly clustered around the 72Hz target throughout the run. ¿¿¿¿¿¿¿ 32df077fa8bbb82b5374f4992fe0114abd8167f7

TABLE III
FRAME RATE PERFORMANCE SUMMARY

State	**Frame Rate (fps)**	**Notes**
Overall Mean	72.2 ± 0.7	Stable Performance
5th–95th Percentile	$70.9 - 73.1$	Includes initialization variance

¡¡¡¡¡¡¡ HEAD Frame rates remained tightly clustered around the 72Hz target during the run, indicating stable performance under the tested load.
=======

4) Calibration Accuracy and Drift: Automated spatial anchor calibration achieved high accuracy under stationary baseline conditions. During the 85.1-minute 3-headset run, the client headset that logged non-zero calibration error reported a mean error of 1.21 ± 0.90mm with a maximum of 5.78mm. This remains within Reimer et al.'s [5] $<$ 10mm safety-critical reference threshold, supporting the feasibility of anchor-based co-location.

TABLE IV
CALIBRATION ACCURACY (HEADSET WITH NON-ZERO LOGGING)

Metric	**Mean (mm)**	**SD**	**Max**
Error	1.21	0.90	5.78

Calibration drift was minimal over the full 85.1-minute 3-headset run, with maximum error remaining well below the 10mm reference threshold.

B. RQ2 & RQ3: Collaboration Effectiveness

1) Task Performance: While specific WIM interface metrics were not collected in this technical validation phase, the system successfully supported 3-user co-located interaction for an 85.1-minute run without technical failure. Under stationary baseline conditions, the shared coordinate system remained within single-digit millimeter error on the headset reporting non-zero calibration error.

C. RQ4: Long-Term Performance Stability

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1) Thermal Performance: Device temperature monitoring revealed consistent thermal buildup patterns. Headsets reached a maximum temperature of 52.6°C.

¡¡¡¡¡¡¡ HEAD Temperature correlated weakly with performance (frame rate vs. temperature: $r = 0.14$).

Despite significant thermal buildup, the system maintained stable performance (72fps and calibration error remaining well below the 10mm threshold) throughout the extended sessions, demonstrating the Quest 3's effective thermal management for long-duration training. ======= Temperature correlated weakly with performance metrics:

- Frame rate vs Temp: $r = 0.14$ (Weak positive correlation)

Despite significant thermal buildup, the system maintained stable performance throughout the extended run, demonstrating the Quest 3's effective thermal management for long-duration training. ¿¿¿¿¿¿¿ 32df077fa8bbb82b5374f4992fe0114abd8167f7

VII. Conclusion

¡¡¡¡¡¡¡ HEAD This study demonstrates that a consumer-grade, wireless 3-user Meta Quest 3 setup can meet selected technical benchmarks relevant to co-located training in a controlled testbed.

A. Key Findings

The technical validation (RQ5) confirmed that the 3-user co-located system met the evidence-based benchmarks used as reference targets in our testbed at the *mean* level. Network latency averaged 58.2 ± 21.3ms, remaining below Van Damme et al.'s RTT target [4] for good collaborative quality of experience, although tail latency exceeded the 75ms target at P95. Under stationary baseline conditions, calibration error remained well below Reimer et al.'s $<$ 10mm reference threshold [5] (mean 1.21 ± 0.90mm; max 5.78mm on the one headset that recorded non-zero values). Frame rate remained stable around the selected 72Hz target (72.2fps) throughout the extended run. Regarding long-term stability (RQ5), thermal analysis showed substantial warming (max 52.6°C) while performance remained stable.

B. Practical Implications

The system demonstrates significant cost-effectiveness. At approximately $2,000 in hardware cost for three Meta Quest 3 headsets plus consumer networking equipment, it enables a portable testbed for co-located collaborative VR without dedicated VR-room infrastructure. Wireless operation offers deployment flexibility by eliminating cable management complexity, enabling training in authentic physical environments rather than dedicated VR rooms. Additionally, the measured calibration error under stationary baseline conditions indicates substantial safety margin relative to the 10mm threshold used as a reference. ======= This study validates consumer-grade wireless VR headsets for professional co-located training applications, demonstrating that Meta Quest 3 systems achieve enterprise-grade performance benchmarks at 15–20% of traditional costs.

C. Key Findings

The technical validation (RQ1) confirmed that the 3-user co-located system met the evidence-based requirements used as reference targets in our testbed. Network latency averaged 58.2 ± 21.3ms, remaining within Van Damme et al.'s threshold [4] for good Quality of Experience. Under stationary baseline conditions, calibration error remained well below Reimer et al.'s $<$ 10mm reference threshold [5] (mean 1.21 ± 0.90mm; max 5.78mm on the headset reporting non-zero calibration error). Furthermore, frame rates remained stable around the native 72Hz target (72.2fps) throughout the extended run. Regarding long-term stability (RQ4), thermal analysis revealed substantial warming (max 52.6°C) while performance remained stable.

D. Practical Implications

The system demonstrates significant cost-effectiveness. At approximately $1,500 total hardware cost (3× Quest 3 headsets) plus consumer networking equipment, it delivers 98% of enterprise VR room capabilities at just 15–20% of typical costs, democratizing access to high-fidelity collaborative VR training. Wireless operation offers deployment flexibility by eliminating cable management complexity, enabling training in authentic physical environments rather than dedicated VR rooms. Additionally, achieving single-digit millimeter calibration error under stationary baseline conditions provides substantial safety margins for collision prevention in co-located training scenarios. ¿¿¿¿¿¿¿ 32df077fa8bbb82b5374f4992fe0114abd8167f7

E. Limitations

While mean latency was acceptable, occasional spikes indicate the need for robust error handling in production environments to ensure network stability. Additionally, this study validated 3-user configurations; scaling to 4 or more users may introduce network congestion, potentially requiring additional infrastructure to maintain performance. ¡¡¡¡¡¡¡ HEAD Because the drift evaluation (RQ2) was conducted with stationary headsets, the reported millimeter-scale alignment should be treated as a best-case baseline and does not, by itself, guarantee collision safety under active user motion.

F. Future Work

1) Scalability and Infrastructure: Future research should systematically evaluate 4-, 5-, and 6-user configurations to determine practical upper limits for scalability. In addition to networking load, these studies should document the infrastructure assumptions required for repeatable low-latency local sessions, such as channel selection, and contention from other nearby devices.

2) Comparative Training Evaluation: To move from technical feasibility to training relevance, controlled studies should compare training outcomes between co-located VR, remote VR, and traditional instruction. For collaborative medical training in particular, structured pilot studies with medical students and clinical staff in short, repeatable scenarios can test whether co-location improves communication, role clarity, and shared situational awareness.

3) Teamwork, Safety, and Motion-Robustness: Future prototypes should incorporate evaluation measures aligned with teamwork and safety objectives, including task completion time, error rates, observer-assessed teamwork/communication rubrics, subjective workload, and explicit safety measures such as near-miss events and minimum spacing between users. Importantly, calibration drift should be re-evaluated under active motion to quantify how alignment error impacts both perceived realism and collision risk.

4) Physical Realism and MR Integration: Purely virtual training remains limited by reduced sensory fidelity. A promising direction is to combine co-located VR with physical props, such as real tools, reference surfaces, or simple mannequins, so that key actions are grounded in touch, while retaining the flexibility and repeatability of virtual content.

5) Spatial Awareness and Coordination Aids: Finally, future work should explore spatial awareness and coordination aids tailored to co-located clinical workflows. This includes testing external-view interfaces, such as world-in-miniature or map-style views for role-based coordination, as well as procedure-specific cues that help users anticipate teammates' movements in tight spaces. These additions should be evaluated not only for usability, but also for whether they reduce collision risk without distracting from the primary task.

G. Concluding Remarks

This research demonstrates that current-generation VR hardware has matured to the point of viability for professional training applications. By systematically validating performance against evidence-based benchmarks, we provide a methodological framework for future co-located multi-user VR system development. The combination of affordable standalone headsets, robust wireless networking, and sophisticated spatial tracking enables a new practical foundation for portable, collaborative VR training. =======

H. Future Work

Future research should systematically evaluate 4-, 5-, and 6-user configurations to determine practical upper limits for scalability. It is also crucial to conduct controlled studies comparing training outcomes between co-located VR, remote VR, and traditional methods to assess learning effectiveness.

I. Concluding Remarks

This research demonstrates that current-generation consumer VR hardware has matured to the point of viability for professional training applications. By systematically validating performance against evidence-based benchmarks, we provide a methodological framework for future co-located multi-user VR system development. The convergence of affordable standalone headsets, robust wireless networking, and sophisticated spatial tracking enables a new paradigm of portable, collaborative VR training.

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Individual Contributions

¡¡¡¡¡¡¡ HEAD This project was completed as part of an academic course on industrial VR applications. The following describes the individual contributions of each team member to the paper and development of a 3-user co-located VR system prototype.

Jon F. Jakobsen

Jon F. Jakobsen focused on the core system architecture and spatial alignment implementation. He integrated Meta's OVR Colocation Discovery API to enable spatial anchor-based alignment and developed the `ColocationManager` for precise camera rig alignment and calibration error tracking. Additionally, he implemented the `MetricsLogger` system for performance data collection and configured the network infrastructure. He also contributed significantly to the technical documentation and overall system architecture design.

Kasper V. Nielsen

Kasper V. Nielsen was primarily responsible for session integration and scene coordination. His work included creating demo scenarios to showcase colocation capabilities and validating the spatial anchor sharing and discovery processes. Furthermore, he contributed to system integration and the development of the interactive scenes. ======= This project was completed as part of an academic course on industrial VR applications. The following describes the individual contributions of each team member to the development and demonstration of a 3-user co-located VR system prototype.

Jon F. Jakobsen

Jon F. Jakobsen focused on the core system architecture and spatial alignment implementation. He integrated Meta's OVR Colocation Discovery API to enable spatial anchor-based alignment and developed the `ColocationManager` for precise camera rig alignment and calibration error tracking. Additionally, he implemented the `MetricsLogger` system for performance data collection and configured the network infrastructure, including the Photon Fusion integration. He also contributed significantly to the technical documentation and overall system architecture design.

Kasper V. Nielsen

Kasper V. Nielsen was primarily responsible for the networking implementation and scene synchronization. He implemented the networking infrastructure using Photon Fusion and developed the `NetworkBehaviour` components required for synchronized objects. His work included creating demo scenarios to showcase colocation capabilities and validating the spatial anchor sharing and discovery processes. Furthermore, he contributed to system integration and the development of the interactive scenes. ¿¿¿¿¿¿¿ 32df077fa8bbb82b5374f4992fe0114abd8167f7

Sebastian M. H. Jensen

Sebastian M. H. Jensen led the research and documentation efforts. He conducted the systematic literature review and evidence synthesis that grounded the project's requirements. He also developed demo scenarios to showcase colocation functionality and configured the data collection protocols and metrics tracking. His contributions extended to documenting system capabilities and technical requirements, as well as writing the Introduction and State of the Art sections.

Sebastian P. H. Pedersen

¡¡¡¡¡¡¡ HEAD Sebastian P. H. Pedersen focused on experimental validation and data analysis. He assisted with the experimental setup and data collection sessions, subsequently analyzing the performance metrics and thermal data. He contributed to the design of the "Shooting Game" scenario and validated system stability during long-duration tests. His writing contributions focused on the Results and Conclusion sections. ======= Sebastian P. H. Pedersen focused on experimental validation and data analysis. He assisted with the experimental setup and data collection sessions, subsequently analyzing the performance metrics and thermal data. He contributed to the design of the "Shooting Game" scenario and validated system stability during long-duration tests. His writing contributions focused on the Results and Discussion sections. ¿¿¿¿¿¿¿ 32df077fa8bbb82b5374f4992fe0114abd8167f7

Shared Responsibilities

All team members contributed equally to the system architecture design, integration decisions, hardware setup, and network configuration. The team also collaborated on documentation, technical writing, paper editing, and the preparation and delivery of the final presentation.

Declaration

We declare that all team members participated actively in this project and contributed substantially to its successful completion. The work presented is original and was conducted in accordance with academic integrity standards.

We further acknowledge the use of Generative AI tools throughout the project lifecycle. These tools were utilized for programming assistance, data analysis, and the retrieval of relevant materials and documentation to support the understanding and resolution of technical challenges. ¡¡¡¡¡¡¡ HEAD
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